

Yafira Martinez

Design Engineer / Creative Technologist

interactive systems, interfaces, and experimental technologies

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EXPERIENCE

Teaching and Academic Support

Graduate Assistant — Intro to Wearables, NYU (IMA)

Jan 2026 – Present

Adjunct Instructor: Daniel Ryan Johnston

- Supporting instruction in wearable and e-textile design, focusing on body-centered interaction, gesture, and non-verbal communication.
- Assisting students with hands-on projects with soft circuits and Adafruit Gemma M0, using Arduino code with light and motion sensors to prototype expressive wearables.

Graduate Assistant — Creative Computing, NYU (IMA)

Sept 2025 – Dec 2025

Professor: Daniel Shiffman

- Supported instruction and mentored students in creative coding through office hours, workshops, and assignment feedback.

Research Assistant — Computational Text, NYU (ITP/IMA)

May 2025–Aug 2025

Professor: Daniel Shiffman

- Assisted in developing the Computational Text course syllabus by identifying tools, artist projects, and frameworks in procedural text, data ethics, and machine learning.
- Researched and curated readings and case studies bridging procedural text, data ethics, and machine learning for creative coding contexts.

Research & Open Source

Machine Learning Researcher & Web Contributor — ml5.js

June 2025 – Present

- Developing ml5.transcribe() and ml5.embeddings(), expanding access to browser-based transformer models (Whisper, Moonshine) for creative coding.
- Refining interaction design, inference performance, and web tooling to support real-time creative ML and contributor usability.

Accessibility Researcher — Making Fashion Accessible, Vertically Integrated Projects (VIP), NYU Ability Lab

Oct 2025–Present

- Researching accessible fashion technologies using knitting machines and AYAB to support inclusive, open-source garment production.
- Contributing to adaptable textile workflows centered on accessibility and human-centered design.

Development & Community

Lead Web Developer — Equitable Syllabus Project, NYU (ITP/IMA)

Nov 2024 – Present

- Designing and implementing accessible, user-centered interfaces using React.js and Figma.
- Integrating and migrating project data while maintaining centralized contribution tracking (Google Sheets) to improve usability and sustainability.

Web Developer & Designer — Useless Machines Archive

Mar 2025 – Aug 2025

Professor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, focusing on client-side data fetching, content organization, and responsive interface design.

Camp Counselor — ITP Camp, New York University

June 2025 - July 2025

- Supported a 150+ person creative technology community through workshops, project mentorship, and hands-on troubleshooting across hardware, software, and fabrication during a two-week intensive.

DESIGN

UX/UI Design, Interaction Design, Responsive Design, Prototyping, Wireframing, Design Systems, Digital Accessibility, Information Architecture, User Research, Figma, Adobe Suite

SOFTWARE

JavaScript, HTML/CSS, React, Next.js, Node.js, APIs/REST APIs, Web Serial, p5.js, ml5.js, Python, CircuitPython, Creative Coding, Git

HARDWARE

Physical Computing, E-Textiles, Soft Circuits, Wearables, Soldering, Fabrication, Microcontrollers, Arduino IDE, Raspberry Pi, Adafruit, Fritzing, Laser Cutter, Knitting Machine

CERTIFICATIONS

Artificial Intelligence,

Samsung Innovation Campus
November 2022 – May 2023

Human Computer Interaction,

Interaction Design Foundation
February 2022

Google UX Design, Coursera

July 2021 – December 2021

EDUCATION

Master of Professional Studies,
Interactive Telecommunications
Program (ITP)

New York University –
Tisch School of Arts
Expected: May 2026

Bachelor of Science,
Computer Science

City University of New York -
Lehman College
December 2020

SELECTED WORK & FEATURES

Guardians of the Groceries (G.O.T.G.), — PlayfulNYC Hackathon 2026, Parsons School of Design April 2026

- Built the hardware and software layer of a handheld ESP32-based device that pairs with a live web registry to document NYC bodega cats. It captures a photo, uploads it to Supabase via WiFi, and logs each cat's alter ego, hidden powers, and cuteness rating. **Awarded** ☆ **Best use of Software** ☆

Open Hardware Summit Fellow — Open Source Hardware Association, Berlin, Germany Feb 2026 - present

- Selected fellow exhibiting NYU ITP thesis "The Soft Computer" and other experimental open-source hardware projects.

Pocket Oracle of Time — Horological Society of New York (HSNY) Feb 2026 - April 2026

- Exhibited a poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics and tactile design.

MIT Hard Mode Hackathon, AI x Hardware 2026 — MIT Media Lab Mar 2026

- Built *enoki*, a focus companion for communal accountability — a React Native app using Mentra smart glasses and Claude Vision AI paired with a desktop companion device using Raspberry Pi, Sseed sensors, and motors, both designed to work together and visualize focus as a mushroom that grows when on task and wilts when distracted.

Moon Pocket — Pedestal Playtime: Art Toy & Tech Toy Open Mic, NYC Resistor Jan 2026

- Showcased a connected device built with an Arduino Nano 33 IoT that fetches real-time moon and tide data from public web APIs and renders it on a round TFT display through touch-based interaction.

Tendertronics: Analog Circuit Performance — NYU ITP's NIME Show, Littlefield Dec 2025

- Performed an analog instrument exploring tactile, minimal interfaces and embodied sound interaction through handcrafted circuitry.

Lightning Talk Speaker — Exploring Creativity in the Age of AI, NYU Global Show & Tell Oct 2025

- Presented *Open Tools, Soft Interfaces*, discussing contributions to ml5.js and open-source machine learning, and framing creativity in AI through openness, design as dialogue, and soft, human-centered interfaces.

UN Open Source Week Hackathon, United Nations Headquarters June 2025

- Participated in a global hackathon focused on open-source solutions for child-centered climate emergency response; contributed to the Ahead of the Storm challenge using geospatial data.

The Internet Book — Annual publication on the poetic web May 2025

- Featured for electrocute.io, a collection of soft-tech and poetic computing experiments, in a curated directory of designers, developers, writers, and digital artists.

Static Chic — Volt Couture: Wearables + E-textiles Open Mic, NYC Resistor May 2025

- Presented *Static Chic*, a 3D-printed inductive handbag that wirelessly lights up embedded LEDs. Part of a wearables and e-textiles open mic curated by Kari Love and Liza Stark.

Rejects: A Benefit Auction, CultureHub May 2025

- Contributed three oddities from my personal archive—*Gothic Teacrypt*, *Orbit Offtrack*, and *Pixel Shell*—to Rejects, a curated auction of discarded prototypes celebrating the charm of creative misfires.

Soft Glitch — eTextile Swatch, Radiant Fibers Exhibition, NYU Clive Davis Gallery Jan 2025 - Mar 2025

- Exhibited an e-textile swatch combining punch needle, chaotic yarns, and electroluminescent thread to explore digital disruption through texture and light.

MIT Reality Hack 2025 — Massachusetts Institute of Technology Jan 2025

- Implemented sound and haptic feedback for *HangarPro VR*, a virtual reality training simulation using Cognitive3D to enhance pre-flight safety through visual interaction analysis for aerospace exploration.